for checking out new software, not clicking on the big-flashing "DOWNLOAD NOW!!!!!" button in your browser and not falling for each and every pop-up while browsing shady sites were sufficient means towards keeping your PC infection free. However, infection vectors are getting numerous, both in number and in virality. Moreover, with powerful state-actors' tools getting leaked onto the internet, "hackers" have it easier now. The flavour of the season is ransomware which lies dormant for a while encrypting your important files before showing you that nasty pop-up asking you to transfer a couple of bitcoins in order to save your files. Since attack vectors are evolving so rapidly, age-old wisdom can only do so much for you, it's better to have seasoned professionals do all the dirty work for you. Professional anti-virus solutions will ensure you're better off in this new age of AI.

NEED TO UPGRADE EVERY X YEARS

This is simply chalked up as marketing mumbo jumbo. On an average, personal computers are upgraded at approximately 5-6 years. And if you've picked the right components, then you can push your computer to perform for a few more years. Back when Intel's Sandy Bridge CPUs came out, they could easily beat Intel's 6-core Core i7-980X in quite a few benchmarks. And the fact remains that if you have a decent graphics card, say an NVIDIA GTX 1070, then you don't need to upgrade your CPU yet. A minor overclock on the Sandy Bridge Core i7-2600K will still give you good gaming performance compared to the recently launched Core i7-8700K.

The simple rule of thumb is to analyse what kind of tasks you perform with your CPU and then upgrade the hardware that results in the most gain. If you game a lot, focus on the graphics card. If you compile massive executables, then a processor is on order, unless you can use GPGPU libraries to push the compile task on the GPU rather than the CPU. And if you multi-task a lot, then more RAM is what you need.

ANYTHING "GAMING" HELPS!

There was an era when you needed hardware specific to get that little edge. Earlier "gaming" hardware had built-in features to help with different gaming scenarios. Now-a-days, the "gaming" label is more of a gimmick with most devices. Let's take a gaming mouse as an example, you can have 10-15 different buttons and awfully weird physical configurations. However, mice are ridiculously subjective and most professional gamers tend to use simple mice such as the Zowie FK1 which is a 3-button mouse with two additional buttons.

If we are to look at a different piece of hardware, such as a motherboard with Killer's NIC, then it was only recently that they've managed to work out the kinks. Killer NICs were known to drop connection speeds randomly and the older software would sometimes prioritise the wrong software in its QoS system which would then require a reset to fix.

If you want to get a "gaming" hardware, then read up on the unique gaming related features that are included and then research to see if it really makes a difference. Blindly buying into gimmicks only makes it worse for the ecosystem as a whole since manufacturers would focus more on gimmicks and less on innovation.

THE HUMAN EYE CAN'T SEE BEYOND 24 FPS

This has long been debunked. The human eye has no fixed frame rate. Our eyes have evolved biologically to discern movement as and where it happens. And we notice motion better in our peripheral vision. For this motion to register, the temporal frequency has to be between 7-13 Hz. After that, persistence of vision sets in and we see parts of each image rather than the entire image. Going by the fact that we can easily perceive the difference in motion, our eyes can notice the difference between 30 and 60 Hz. So a 60 Hz monitor is more than sufficient. However, we can easily see the difference between a 60 Hz monitor and a 120 Hz monitor and even a 144 Hz monitor. The images become ridiculously smooth. However, there is the question of display technologies. If you were to use a crappy VA panel with a driver pushing 120 Hz, you'll have a poor experience because you'll still see ghosting. What we're trying to say is that if you are going for a 120 Hz monitor or better, you should go for a good quality panel that is capable of making the most of the faster refresh rate. Also, it goes without saying that you should have a good graphics card for the higher FPS.

ESD WILL KILL YOUR COMPONENTS

Electrostatic Discharge has often been a cause for worry when dealing with sensitive electronic components. This is still true to quite an extent, however, most components these days have built-in ESD protection mechanisms with multiple capacitors to and fuses to prevent a high discharge from overloading any of the components. Usually, you'll find such mechanisms for the USB and LAN ports on a motherboard.

The simple fact is that you no longer need ESD protection in the form of an anti-static wrist strap or floor mat. Nor do you need to get rid of your floor carpet before handling the innards of your PC. However, it wouldn't hurt to do so. Honestly speaking, most of us in the Test Centre haven't shorted out any CPU / GPU or motherboard for ages and we work in a relatively cool environment. Moreover, some of the folks in the Test Centre have a affinity towards static charge build up, and yet, there have been no mishaps.

YOU NEED A CAD/GAMING /EDITING PC

The average PC can run everything. Given how PC hardware have evolved to become really powerful and when you consider that most hardware is practically similar except for a few minor changes, what you have is sufficient for most situations. Let's say you want to start playing around with CAD or video editing, then conventional wisdom would require you to invest in specialised hardware, i.e. a workstation CPU such as Intel's Xeon or AMD's EPYC and a workstation GPU such as NVIDIA's Quadro or AMD's RadeonPro. However, such hardware only come into play when you need to build a production quality system. The average PC is quite capable of handling these tasks, you'll just end up spending more time to get the job done. d